Mohammed Mahdi Nsr

AI Engineer

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GitHub

Summary

AI Engineer focused on developing intelligent AI systems powered by LLMs, RAG, and Multi-Agent architectures, with expertise in Machine Learning, Deep Learning, NLP and Computer vision. Experienced in transforming cutting-edge AI research into scalable, production-ready solutions using LangChain, LangGraph, CrewAI, Hugging Face, and vector databases

Experience

IT Specialist & System Administrator, *Military Service* 01/2025 – 03/2026

AI & Data Science Trainee, *Prodigy InfoTech* 07/2024 – 09/2024

Designed and implemented two end-to-end machine learning projects as part of AI & Data Science training:

- Movie Recommender System (Content-based filtering for movie suggestions)
- Sentiment Analysis on Amazon Reviews (NLP classification of customer feedback)

AI & Data Science Trainee, *Information Technology Institute (ITI)* 08/2023 – 09/2023

Completed intensive training program in Artificial Intelligence and Data Science. Gained strong understanding of core AI concepts including machine learning fundamentals, data preprocessing, model evaluation, and basic neural networks. Developed hands-on experience in applying AI techniques to real-world problems. Built two end-to-end projects demonstrating practical implementation of learned concepts.

Education

Bachelor's degree in Computer Science, *Beni Suef university* 2020 – 2024

Grade : Very Good Faculty of computers and Artificial intelligence Beni Suef university

Skills

Programming Principles

Problem solving, OOP, Database (Sql server), Data structures, Algorithms, Math for AI Python

Agentic AI & LLMs

Agentic AI & LLMs, AI Agents, GenAI, HuggingFace, LangChain, LangGraph, CrewAI, RAG, Fine-tuning, VLMs (OCR), Vector database (Chroma, Faiss, Pinecone &Quadrant)

AI & ML Concepts

EDA, Data Preprocessing, Text Preprocessing, NLTK & Spacy Transformers, Scikit-learn, Pandas, NumPy, ML algorithms, PyTorch, TensorFlow, Keras

Deployment

APIs (FastAPIs), Git & GitHub, Docker, AWS, Google Cloud Platform (GCP)

Projects

Egyptian-Legal-Research-Agent, 05/2026 – 05/2026

Multi-Agent Legal Research AI System for Egyptian Civil Law

- Built a Multi-Agent Legal Research AI System for Egyptian law Civil Law using LangGraph and RAG architecture.
- Implemented specialized agents for orchestration, Retrieval, Tavily-web search, Reranking,Critique, and Answer generation to automate complex legal research workflows. Improved retrieval relevance and factual accuracy using Query Rewriting, Multi-Query Retrieval, Corrective RAG, and Cross-Encoder Reranking, reducing hallucinations by 93% and enabling the generation of highly reliable, evidence-grounded legal research reports. Exposed the solution through a Flask API, containerized it with Docker, and deployed it on AWS to ensure scalable, maintainable, and production-ready operations

LLM Fine-Tuning for Docker Command Generation 04/2026 – 04/2026

Fine-tuned Hugging Face Qwen2.5-Coder-1.5B-Instruct using LoRA (Low-Rank Adaptation) to generate Docker CLI commands from natural language instructions. Implemented and evaluated supervised fine-tuning (SFT) workflows using LlamaFactory, Docker, and Linux-based environments, building an instruction-following NLP system capable of producing valid Docker commands with optimized inference efficiency. Released the model publicly, resulting in 5+ downloads and real-world usage.

Multilingual Text Classification System 03/2026 – 04/2026

Designed and implemented an end-to-end multilingual NLP system for detecting and classifying Arabic and English text using machine learning techniques.

- Arabic Text Classification: Built an Arabic text classification pipeline including data preprocessing, EDA, TF-IDF feature extraction, model training, and evaluation. Performed exploratory data analysis covering class distribution, text length analysis, frequent word extraction, and word cloud visualization. Trained and evaluated the model using Gaussian Naïve Bayes with classification reports and confusion matrices.
- English Text Classification: Developed a machine learning model to automatically distinguish between Arabic and English text. Combined Arabic and English datasets to improve generalization and overall model performance.
- Full System Implementation: Implemented the final solution as a structured Python project with a modular architecture, including separate components for: Language Detection / Arabic Text Classification / English Text Classification / Main Application Workflow

Languages

Arabic
Native/Bilingual

English
Proficient